TECHNICAL & CODING

100 PYQ-Style Questions — Practice Set 2

Placement & Competitive Exam Practice

Included	100 Distinct PYQ-Style Questions • Practice Set 2
Useful For	TCS NQT • Cognizant • Infosys • Wipro • Accenture • Bosch • Campus Placements
Topics Covered	C • C++ • Java • Python • Data Structures • DBMS & SQL • JavaScript • OS • Networks • OOPs
Extra	Answer Key on Each Page • Practice Set 2 • 1 Page per Language

Structure: 10 Core Programming Languages & Technical Domains. Set 2 contains 100 unique questions. Each language contains exactly 10 authentic MCQs with an Answer Key box at the bottom of the page.

TECHNICAL ASSESSMENT • C PROGRAMMING

Practice Set 2 • Section 1 of 10 • 10 Questions with Answer Key

Q1. What is the output of the bitwise operation printf("%d", 5 >> 1) in C?

A) 10
B) 2
C) 2.5
D) 1

Q2. Which of the following is NOT a legally valid variable identifier in C?

A) _total_count
B) 1st_rank
C) sumOfMarks
D) var2

Q3. What is the typical memory size of a pointer variable on a 64-bit operating system architecture?

A) 2 bytes
B) 4 bytes
C) 8 bytes (64 bits)
D) 16 bytes

Q4. Which C standard library function is used to compare two null-terminated strings?

A) strcat()
B) strcpy()
C) strcmp()
D) strlen()

Q5. What does declaring a local variable as static inside a function accomplish in C?

A) Exports it globally
B) Preserves its value across successive function calls
C) Makes it read-only
D) Places it on stack

Q6. What is the evaluated result of the integer division expression 7 / 2 in C?

A) 3.5
B) 3
C) 4
D) 3.0

Q7. Which operator among the following exhibits the highest precedence in C?

A) + (Addition)
B) * (Multiplication)
C) () (Parentheses / Function Call)
D) == (Equality)

Q8. In C programming, what is the default starting index of all arrays?

A) 1
B) 0
C) -1
D) Configurable

Q9. Which preprocessor directive is used to define symbolic constants and macros in C?

A) #include
B) #define
C) #typedef
D) #pragma

Q10. Which function reads formatted input from the standard input stream (stdin) in C?

A) printf()
B) scanf()
C) puts()
D) getchar()

ANSWER KEY & SOLUTIONS — C PROGRAMMING (SET 2)

Q1: B • Q2: B • Q3: C • Q4: C • Q5: B

Q6: B • Q7: C • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • C++ PROGRAMMING & OOPS

Practice Set 2 • Section 2 of 10 • 10 Questions with Answer Key

Q1. Which standard stream object is used for unbuffered error output in C++?

A) std::cout
B) std::cin
C) std::cerr
D) std::clog

Q2. What is Function Overloading in C++?

A) Defining multiple functions with identical name but different parameter lists in the same scope
B) Overriding virtual methods
C) Operator overload
D) Recursive call

Q3. What is the default member access specifier in a C++ class if none is explicitly specified?

A) public
B) protected
C) private
D) internal

Q4. What does the 'this' pointer represent inside an active non-static C++ member function?

A) Address of base class
B) Pointer holding the memory address of the invoking object instance
C) Class descriptor
D) Static table

Q5. Which trio of keywords implements exception handling mechanisms in C++?

A) try, catch, throw
B) attempt, catch, raise
C) try, except, finally
D) begin, catch, throw

Q6. What is the correct parameter signature for a standard C++ Copy Constructor?

A) ClassName(ClassName obj)
B) ClassName(const ClassName& obj)
C) ClassName(ClassName* obj)
D) ClassName(void)

Q7. Which STL container is internally implemented as a Doubly Linked List in C++?

A) std::vector
B) std::deque
C) std::list
D) std::forward_list

Q8. Can a static member function access non-static instance variables directly in C++?

A) Yes, always
B) No, because static member functions do not possess a 'this' pointer
C) Only if public
D) Only if mutable

Q9. Which operator is the Scope Resolution Operator in C++?

A) . (dot)
B) -> (arrow)
C) :: (double colon)
D) : (single colon)

Q10. What hint does the 'inline' keyword provide to the C++ compiler?

A) Allocate variable in CPU cache
B) Substitute function call with actual body code to eliminate call overhead
C) Launch in separate thread
D) Mark as deprecated

ANSWER KEY & SOLUTIONS — C++ PROGRAMMING & OOPS (SET 2)

Q1: C • Q2: A • Q3: C • Q4: B • Q5: A

Q6: B • Q7: C • Q8: B • Q9: C • Q10: B

TECHNICAL ASSESSMENT • JAVA CORE & OOPS

Practice Set 2 • Section 3 of 10 • 10 Questions with Answer Key

Q1. Which class serves as the ultimate cosmic superclass for every class in Java?

A) java.lang.Class
B) java.lang.Object
C) java.lang.System
D) java.lang.Base

Q2. Which runtime memory area in the JVM is globally shared among all executing threads?

A) Thread Stack
B) Program Counter Register
C) Heap Memory
D) Native Method Stack

Q3. What is the default initial value of an uninitialized boolean instance field in Java?

A) true
B) false
C) null
D) 0

Q4. Which Java keyword explicitly prevents a method from being overridden by any subclass?

A) static
B) final
C) abstract
D) synchronized

Q5. What is the exact memory size and signed bit range of the 'int' primitive data type in Java?

A) 2 bytes (16 bits)
B) 4 bytes (32 bits)
C) 8 bytes (64 bits)
D) Architecture dependent

Q6. Which interface must a Java class implement to define natural sorting order with Collections.sort()?

A) java.lang.Comparable
B) java.util.Comparator
C) java.lang.Cloneable
D) java.io.Serializable

Q7. What is Autoboxing in the Java language?

A) Automatic conversion of primitive data types into their corresponding wrapper objects
B) Casting double to int
C) Compiling .class files
D) JVM memory defragmentation

Q8. Which class should be chosen when frequent thread-safe string modifications are required in Java?

A) java.lang.String
B) java.lang.StringBuffer
C) java.lang.StringBuilder
D) java.lang.CharBuffer

Q9. Can a constructor in Java declare a return type such as void or int?

A) Yes, void is required
B) No, constructors cannot declare any return type; doing so converts it into a regular method
C) Yes, int status
D) Only in abstract classes

Q10. Which of the following is classified as a Checked Exception in Java?

A) NullPointerException
B) IOException
C) ArithmeticException
D) ArrayIndexOutOfBoundsException

ANSWER KEY & SOLUTIONS — JAVA CORE & OOPS (SET 2)

Q1: B • Q2: C • Q3: B • Q4: B • Q5: B

Q6: A • Q7: A • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • PYTHON PROGRAMMING

Practice Set 2 • Section 4 of 10 • 10 Questions with Answer Key

Q1. What is the evaluated output of len({'a': 1, 'b': 2, 'a': 3}) in Python 3?

A) 3
B) 2 (duplicate key 'a' is updated/overwritten)
C) 1
D) KeyError

Q2. How do you correctly create a completely empty Set object in Python?

A) {}
B) set()
C) []
D) ()

Q3. What list of numbers is produced by the generator list(range(1, 10, 2)) in Python?

A) [1, 3, 5, 7, 9]
B) [1, 2, 3, 4, 5]
C) [2, 4, 6, 8, 10]
D) [1, 9]

Q4. Which keyword is utilized to define an anonymous inline lambda function in Python?

A) def
B) lambda
C) inline
D) arrow

Q5. What does the string method 'hello'.find('z') return in Python when the substring is absent?

A) False
B) None
C) -1
D) ValueError

Q6. What is a Decorator in Python programming?

A) A function that takes another function as an argument and dynamically enhances its behavior
B) GUI formatting utility
C) Class variable
D) Syntax highlighter

Q7. Which built-in standard library module provides Regular Expression capabilities in Python?

A) regex
B) re
C) string
D) pattern

Q8. What is the exact purpose of the 'pass' keyword in Python syntax?

A) Terminate loop
B) A null statement used as a placeholder where code is syntactically required
C) Return None
D) Skip next block

Q9. What is the resulting output of the sequence multiplication [1, 2] * 3 in Python?

A) [3, 6]
B) [1, 2, 1, 2, 1, 2]
C) [[1, 2], [1, 2], [1, 2]]
D) TypeError

Q10. Which statement structure handles runtime errors and exceptions in Python?

A) try ... except ... finally
B) try ... catch ... finally
C) begin ... rescue
D) guard ... catch

ANSWER KEY & SOLUTIONS — PYTHON PROGRAMMING (SET 2)

Q1: B • Q2: B • Q3: A • Q4: B • Q5: C

Q6: A • Q7: B • Q8: B • Q9: B • Q10: A

TECHNICAL ASSESSMENT • DATA STRUCTURES & ALGORITHMS

Practice Set 2 • Section 5 of 10 • 10 Questions with Answer Key

Q1. What is the average and worst-case time complexity of searching an element in an unsorted array?

A) O(1)
B) O(log N)
C) O(N)
D) O(N^2)

Q2. In a circular queue of capacity N, which formula correctly indicates that the queue is completely full?

A) front == rear
B) (rear + 1) % N == front
C) rear == N - 1
D) front == 0

Q3. What is the best-case time complexity of an optimized Bubble Sort when the input array is already sorted?

A) O(N^2)
B) O(N)
C) O(N log N)
D) O(1)

Q4. Which binary tree traversal algorithm recursively visits nodes in the order: Left, Root, Right?

A) Pre-order
B) In-order
C) Post-order
D) Level-order

Q5. What is the auxiliary space complexity of standard iterative Binary Search?

A) O(1)
B) O(log N)
C) O(N)
D) O(N log N)

Q6. Which graph representation consumes O(V^2) space regardless of whether edges are dense or sparse?

A) Adjacency List
B) Adjacency Matrix
C) Edge List
D) Incidence Matrix

Q7. What fundamental property defines a Min-Heap binary tree?

A) Root node holds the minimum key and each parent node is <= its child nodes
B) Root holds max key
C) Left child always > right child
D) Must be a full BST

Q8. Which algorithmic paradigm breaks down problems into non-overlapping subproblems, solves them, and merges results?

A) Dynamic Programming
B) Divide and Conquer
C) Backtracking
D) Greedy Approach

Q9. What is the maximum height of a balanced Binary Search Tree containing N total nodes?

A) O(N)
B) O(log2 N)
C) O(1)
D) O(N^2)

Q10. In a singly linked list with head pointer, what is the time complexity to insert a new node at the head?

A) O(1)
B) O(N)
C) O(log N)
D) O(N^2)

ANSWER KEY & SOLUTIONS — DATA STRUCTURES & ALGORITHMS (SET 2)

Q1: C • Q2: B • Q3: B • Q4: B • Q5: A

Q6: B • Q7: A • Q8: B • Q9: B • Q10: A

TECHNICAL ASSESSMENT • SQL & DATABASE MANAGEMENT (DBMS)

Practice Set 2 • Section 6 of 10 • 10 Questions with Answer Key

Q1. Which SQL statement is used to completely remove a table along with its schema and indexes from the database?

A) DELETE TABLE
B) REMOVE TABLE
C) DROP TABLE
D) TRUNCATE TABLE

Q2. What properties must a relational database column fulfill to serve as a PRIMARY KEY?

A) Must allow duplicates
B) Must contain unique values and NOT permit NULL values
C) Must reference foreign table
D) Must be integer type

Q3. Which SQL aggregate function computes the total arithmetic sum of numeric values in a column?

A) TOTAL()
B) COUNT()
C) SUM()
D) AVG()

Q4. What does the DISTINCT keyword achieve when added to a SQL SELECT query?

A) Sorts rows alphabetically
B) Eliminates duplicate rows from the final query result set
C) Groups records
D) Filters NULL values

Q5. What requirement must a relational table satisfy to achieve Second Normal Form (2NF)?

A) Must be in 1NF and contain no partial dependency of any non-prime attribute on candidate keys
B) No transitive dependencies
C) Clustered index
D) Foreign key required

Q6. Which SQL operator performs pattern matching using wildcard characters (% and _)?

A) IN
B) LIKE
C) BETWEEN
D) REGEXP_ONLY

Q7. How does a Clustered Index physically organize records on disk within a relational database?

A) Physically sorts and stores the actual table data rows in the index key order
B) Stores pointers in external tree
C) Duplicates data in RAM
D) Disables range queries

Q8. What is a Database View?

A) Physical duplicate copy of data
B) A virtual table based on the dynamic result-set of a pre-written SQL SELECT statement
C) Hardware partition
D) Stored backup

Q9. Which SQL clause sorts the returned records in ascending or descending order?

A) GROUP BY
B) SORT BY
C) ORDER BY
D) ARRANGE BY

Q10. What is the primary effect of issuing a ROLLBACK command in SQL transaction control?

A) Permanently commits data
B) Reverts all uncommitted database changes made during the active transaction
C) Drops tables
D) Reboots database engine

ANSWER KEY & SOLUTIONS — SQL & DATABASE MANAGEMENT (DBMS) (SET 2)

Q1: C • Q2: B • Q3: C • Q4: B • Q5: A

Q6: B • Q7: A • Q8: B • Q9: C • Q10: B

TECHNICAL ASSESSMENT • JAVASCRIPT & WEB TECHNOLOGIES

Practice Set 2 • Section 7 of 10 • 10 Questions with Answer Key

Q1. Which keyword in modern ES6 JavaScript declares a re-assignable block-scoped variable?

A) const
B) let
C) var
D) static

Q2. What does Boolean('false') evaluate to in JavaScript type coercion?

A) false
B) true (any non-empty string evaluates to truthy)
C) undefined
D) TypeError

Q3. What is Hoisting in JavaScript execution context?

A) Moving variable and function declarations to the top of their containing scope during compilation
B) Event bubbling
C) Garbage collection
D) Asynchronous polling

Q4. Which Array method creates a brand new array populated with the results of calling a function on every element?

A) filter()
B) forEach()
C) map()
D) reduce()

Q5. How is the 'this' value resolved inside an ES6 Arrow Function?

A) Lexically inherited from the enclosing execution context scope
B) Always points to global window object
C) Undefined always
D) Bound to DOM event target

Q6. What is the evaluated output of typeof null in JavaScript due to a historic implementation artifact?

A) 'null'
B) 'undefined'
C) 'object'
D) 'boolean'

Q7. Which Array method appends one or more items to the end of an array and returns the updated array length?

A) unshift()
B) push()
C) pop()
D) shift()

Q8. What does the expression NaN === NaN return in JavaScript according to IEEE-754 specifications?

A) true
B) false
C) undefined
D) TypeError

Q9. Which operator syntax checks whether a specific property key exists in a JavaScript object?

A) 'propName' in object
B) object.hasProperty('propName')
C) object.contains('propName')
D) exists(object.propName)

Q10. What underlying asynchronous architectural primitive powers JavaScript's async/await syntax?

A) Callbacks
B) Promises and Generators
C) Web Sockets
D) Multi-threaded CPU locks

ANSWER KEY & SOLUTIONS — JAVASCRIPT & WEB TECHNOLOGIES (SET 2)

Q1: B • Q2: B • Q3: A • Q4: C • Q5: A

Q6: C • Q7: B • Q8: B • Q9: A • Q10: B

TECHNICAL ASSESSMENT • OPERATING SYSTEMS & LINUX

Practice Set 2 • Section 8 of 10 • 10 Questions with Answer Key

Q1. What is the primary function of the CPU Dispatcher module in an Operating System?

A) Selects next process from ready queue
B) Transfers CPU control to the process selected by the short-term scheduler
C) Allocates RAM pages
D) Compiles code

Q2. What triggers a Page Fault interrupt in virtual memory management?

A) Disk sector failure
B) An active process accessing a virtual memory page not currently mapped in physical RAM
C) CPU overheating
D) Stack overflow

Q3. Which Inter-Process Communication (IPC) mechanism operates as a First-In-First-Out named pipe?

A) Message Queue
B) Shared Memory
C) Named Pipe (FIFO)
D) Unix Domain Socket

Q4. What defines a Zombie Process in Unix/Linux operating systems?

A) A terminated process whose exit status has not yet been collected by its parent process via wait()
B) An orphaned daemon
C) A crashed kernel thread
D) A high-CPU infinite loop

Q5. What does SPOOLING stand for in operating system peripheral management?

A) Simultaneous Peripheral Operations On-Line
B) Sequential Program Output Over Line
C) Shared Process Operating Offline
D) System Page Output On Load

Q6. Which disk head scheduling algorithm is commonly referred to as the Elevator Algorithm?

A) FCFS
B) SSTF
C) SCAN
D) C-LOOK

Q7. What is the classic Banker's Algorithm used for in resource management?

A) Deadlock Avoidance (detecting safe states)
B) Deadlock Recovery
C) Priority Scheduling
D) Paging replacement

Q8. What signal is transmitted to a process when executing kill -9 <PID> in a Linux shell?

A) SIGTERM (Graceful termination)
B) SIGKILL (Forced uncatchable termination)
C) SIGHUP
D) SIGINT

Q9. What is Belady's Anomaly in operating system memory management?

A) Experiencing more page faults when increasing allocated page frames (occurs in FIFO)
B) High disk latency
C) Memory fragmentation
D) Cache overflow

Q10. What differentiates Preemptive CPU scheduling from Non-Preemptive scheduling?

A) In preemptive scheduling, the OS can interrupt and take CPU away from a running process
B) Non-preemptive uses timer interrupts
C) Preemptive does not support priority
D) No difference

ANSWER KEY & SOLUTIONS — OPERATING SYSTEMS & LINUX (SET 2)

Q1: B • Q2: B • Q3: C • Q4: A • Q5: A

Q6: C • Q7: A • Q8: B • Q9: A • Q10: A

TECHNICAL ASSESSMENT • COMPUTER NETWORKS

Practice Set 2 • Section 9 of 10 • 10 Questions with Answer Key

Q1. Which application-layer protocol is responsible for transferring outgoing email between mail transfer agents?

A) POP3
B) IMAP
C) SMTP (Simple Mail Transfer Protocol)
D) HTTP

Q2. What is the standard subnet mask for a traditional Class C IPv4 network block?

A) 255.0.0.0
B) 255.255.0.0
C) 255.255.255.0
D) 255.255.255.255

Q3. Which network hardware device operates at Data Link Layer (Layer 2) to filter and forward frames via MAC addresses?

A) Hub
B) Network Switch
C) Router
D) Repeater

Q4. What is the primary function of DHCP (Dynamic Host Configuration Protocol)?

A) Translate domain names to IP
B) Automatically allocate IP addresses, subnet masks, and default gateways to client devices
C) Encrypt network traffic
D) Monitor packet loss

Q5. Which protocol provides secure encrypted command-line shell access to remote servers over port 22?

A) Telnet
B) FTP
C) SSH (Secure Shell)
D) SNMP

Q6. What is the standard default Maximum Transmission Unit (MTU) size of a standard Ethernet packet frame?

A) 512 bytes
B) 1500 bytes
C) 4096 bytes
D) 64 bytes

Q7. What is the key purpose of Network Address Translation (NAT) in modern networking?

A) Maps multiple private local IP addresses to a single public IP address for internet routing
B) Resolves domain names
C) Filters malware
D) Compresses data

Q8. Which HTTP response status code indicates '404 Not Found'?

A) 200
B) 301
C) 404
D) 500

Q9. What is an IPv4 Directed Broadcast Address used for on an IP subnet?

A) 127.0.0.1 loopback
B) Packet delivery to every host on that specific subnet (all host bits set to 1)
C) Default route 0.0.0.0
D) Multicast address

Q10. Which transport layer protocol operates without establishing connections and has minimal delivery overhead?

A) TCP
B) UDP (User Datagram Protocol)
C) SCTP
D) BGP

ANSWER KEY & SOLUTIONS — COMPUTER NETWORKS (SET 2)

Q1: C • Q2: C • Q3: B • Q4: B • Q5: C

Q6: B • Q7: A • Q8: C • Q9: B • Q10: B

TECHNICAL ASSESSMENT • OBJECT-ORIENTED PROGRAMMING (OOPS)

Practice Set 2 • Section 10 of 10 • 10 Questions with Answer Key

Q1. What does Polymorphism allow in object-oriented software engineering?

A) Bundling data into classes

B) The ability of different classes to respond to the same method invocation in distinct specialized ways

C) Eliminating heap memory

D) Restricting member visibility

Q2. What defines an Abstract Method in object-oriented design?

A) A method declared without implementation that must be implemented by concrete derived subclasses

B) A static helper method

C) A method with empty brackets

D) A final method

Q3. Which SOLID principle mandates that software modules should be open for extension, but closed for modification?

A) Single Responsibility Principle

B) Open/Closed Principle (OCP)

C) Interface Segregation Principle

D) Dependency Inversion Principle

Q4. What is the primary role of a Copy Constructor in object initialization?

A) Allocates static memory

B) Initializes a newly created object as an exact copy of an existing object of the same class

C) Compiles templates

D) Deletes old instances

Q5. What architectural problem arises from Tight Coupling between system classes?

A) Modules become interdependent, making changes in one class break functionality in another

B) Reduced execution speed

C) Stack overflow errors

D) Higher memory consumption

Q6. Which relationship in object modeling describes an 'IS-A' inheritance hierarchy?

A) Inheritance / Generalization

B) Composition

C) Aggregation

D) Dependency

Q7. What is the core benefit of Encapsulation in object-oriented systems?

A) Faster execution loops

B) Protecting internal state integrity by mediating all data access through validated getter/setter methods

C) Multiple inheritance

D) Global access

Q8. What is Operator Overloading?

A) Declaring duplicate operators

B) Customizing how standard built-in language operators behave when applied to user-defined class objects

C) Stack memory leak

D) Type coercion error

Q9. Which GoF design pattern defines an interface for creating objects, letting subclasses decide which class to instantiate?

A) Singleton Pattern

B) Factory Method Pattern

C) Observer Pattern

D) Adapter Pattern

Q10. Can an Interface be directly instantiated using the 'new' operator in Java or C#?

A) Yes, if it has default methods

B) No, interfaces lack complete implementations and can only be instantiated via implementing classes

C) Yes, always

D) Only in static contexts

ANSWER KEY & SOLUTIONS — OBJECT-ORIENTED PROGRAMMING (OOPS) (SET 2)

Q1: B • Q2: A • Q3: B • Q4: B • Q5: A

Q6: A • Q7: B • Q8: B • Q9: B • Q10: B